1891–1941

FREDERICK BANTING

MILESTONES

BECOMES DOCTOR
Switches from divinity to medicine in 1916; receives medical degree from University of Toronto.

MILITARY HERO
Serves in the Canadian Army Medical Corps and is awarded the Military Cross in 1919.

DIABETES RESEARCH
In 1920, explains his idea for isolating insulin to Professor John Macleod at University of Toronto.

NOBEL PRIZE WINNER
Awarded the Nobel Prize in 1923 and shares the money with his assistant Charles Best.

The hormone insulin, isolated by Canadian physician Sir Frederick Banting in 1921, fast became the standard treatment for diabetes.

Frederick Grant Banting studied medicine at the University of Toronto. At the time, reports suggested that diabetes was caused by a lack of insulin—a hormone that controls the metabolism of sugar—but no one knew how to extract it from the pancreas. The trouble was that other digestive enzymes destroyed it. In 1920, Banting read that closing the pancreatic duct kills the acini cells of the pancreas, which make digestive enzymes, but leave the cells that secrete insulin intact.

Banting began by operating on dogs, closing their pancreatic ducts before operating again to extract their insulin. He then injected this into diabetic dogs, and it successfully lowered their sugar levels. In 1922, 14-year-old Leonard Thompson became the first human being to receive this treatment—for which Banting received the Nobel Prize in 1923.

***Banting loved dogs**, and it upset him to use them in his insulin trials.*

"The **greatest joy** in life is to **accomplish …**"

Frederick Banting, 1928